1. Let A be a Wishart $W_n(m, \Sigma)$. For a $k \times n$ matrix M of full row rank, find the characteristic function of MAM^T.

2. Let A be a Wishart $W_n(m, \Sigma)$. Find an unbiased estimator of Σ^{-1}.

3. Let $A \sim W_p(\nu, M)$ follows Wishart distribution with scale matrix M and ν degrees of freedom. Prove that $E(A) = \nu M$. What is the general procedure for matrix valued distributions?

4. Given the data matrix $\begin{bmatrix} 7 & 7 \\ 1 & 9 \\ 7 & 6 \\ 5 & 6 \end{bmatrix}$ specify the Hotelling's T^2 statistics and its distribution. Test $H_0 : \underset{\sim}{\mu} = (7, 11)^T$ at 0.05 significance level.

5. In a study of home-schooling, 10 sixteen-year-olds who have been home-schooled and 20 sixteen-year-olds who attended a grammar school were examined in both English and mathematics. The sample mean vectors and variance matrices for the two tests (English as variable 1 and mathematics as variable 2) for the two groups are given below, where the home-schooled group is denoted by x and the other by y:

$$\bar{\underset{\sim}{x}} = (77.2.62.4)^T,\ \bar{\underset{\sim}{y}} = (72.4, 63.1)^T,\ S_x = \begin{bmatrix} 96.5 & 47.6 \\ 47.6 & 47.1 \end{bmatrix},\ S_y = \begin{bmatrix} 45.6 & 26.4 \\ 26.4 & 26.2 \end{bmatrix}$$

Use Hotelling's T^2 test to test the difference in means at a 5% level of significance. Calculate simultaneous confidence intervals for a $a^T(\mu_x - \mu_y)$ for vectors $a = (1, 0)^T$ and$a = (0, 1)^T$ and interpret your findings.

6. A research company conducts an experiment to compare the efficiency of 5 drugs. These are 20 animals available for the trial and each drug is injected into 4 randomly selected animals. Three animals die during the course of the experiment. The blood samples from the remaining animals are taken and analyzed. The data on blood pH reading from each blood analysis in certain standardized units are given in the following table.

$$\begin{bmatrix} A & B & C & D & E \\ 19 & 7 & 4 & 6 & 6 \\ 11 & 1 & 7 & 6 & 4 \\ 15 & 4 & 7 & 6 & 2 \\ & 4 & & 10 & \end{bmatrix}$$

a) Analyze the data and test if all the drugs are equally effective or not.
b) Consider the contrast $L = \mu_1 + 2\mu_2 - \mu_3 - \mu_4 - \mu_5$.
Test the hypothesis $H_0 : L = 0$ versus $H_a : L \neq 0$ at 95% confidence level.

7. In the study of radiation from microwave ovens, transformed variables.
$X_1 =$ (measured radiation from microwave ovens with doors closed)$^{0.25}$
$X_2 =$ (measured radiation from microwave ovens with doors open)$^{0.25}$.
have a nearly Bivariate Normal distribution. From a sample of 42 ovens,

$$\bar{\underset{\sim}{X}} = \begin{bmatrix} 0.56 \\ 0.60 \end{bmatrix},\ S = \begin{bmatrix} 0.014 & 0.012 \\ 0.012 & 0.015 \end{bmatrix},\ S^{-1} = \begin{bmatrix} 203 & 163 \\ 163 & 200 \end{bmatrix}$$

(a) Conduct a test of the null hypothesis $H_0 : \mu = (0.55, 0.60)^T$ at the 0.05 level of significance. For this test, compute the Hotelling's T^2 statistic directly and by using Wilks' Lambda.
(b) Write an equation for the 95% confidence ellipsoid for the multivariate mean. How far are the vertices of this confidence ellipsoid from its center?
(c) Based on this ellipsoid, what will be the result of testing $H_0 : \mu = (0.7, 0.7)^T$ against $H_a : \mu \neq (0.7, 0.7)^T$ at $\alpha = 0.05$?
(d) Construct simultaneous confidence intervals for the univariate means $\mu_1 = E(X_1)$ and$\mu_2 = E(X_2)$ with the joint coverage probability of at least 95%.
(e) Use these confidence interval to test $H_0 : \mu = (0.7, 0.7)^T$ against $H_a : \mu \neq (0.7, 0.7)^T$ at $\alpha = 0.05$.
(f) Construct simultaneous T^2 intervals for μ_1 and μ_2 and again, test H_0 vs H_a as in (e).

8. A survey commissioned a study of womens nutrition. Nutrient intake was measured for a random sample of 402 women aged 25-50 years. One of the questions of interest is whether women meet the nutritional intake guidelines. The table below shows the recommended daily intake and the sample means for iron and protein:

Variable Recommended	Intake	Sample mean
Iron	15 mg	11 mg
Protein	60 g	66 g

The sample variance-covariance matrix was (rounded) $S = \begin{bmatrix} 30 & 100 \\ 100 & 1000 \end{bmatrix}$

Does a Hotelling's T^2 test shows a significant evidence at 5% level that on an average women fail to meet the nutrition guidelines?

9. Consider the length (in cm) data of female bears at 2, 3, 4, 5 years of age as follows.

Bear	L2	L3	L4	L5
1	141	157	168	183
2	140	168	174	170
3	145	162	172	177
4	146	159	176	171
5	150	158	168	175
6	142	140	178	189
7	139	171	176	175

(a) Obtain the 95% T^2 simultaneous confidence intervals for 4 population mean for length.
(b) Obtain the 95% T^2 simultaneous confidence intervals for the three successive yearly increase in mean length.
(c) Obtain the 95% T^2 confidence ellipse for the mean increase length from 2 to 3 years and the mean increase length from 4 to 5 years.
(d) Construct the 95% Bonferroni confidence intervals for the set containing mean lengths and the three successive yearly increase in mean length.
(e) Compare the 95% Bonferroni confidence rectangle for the mean increase in length from 2 to 3 years about the mean increase in length from 4 to 5 years with the ellipse produce for the same by the T^2 procedure.

10. Given the data $\begin{bmatrix} 3 & 6 & 0 \\ 4 & 4 & 3 \\ * & 8 & 3 \\ 5 & * & * \end{bmatrix}$ with missing components. Use the prediction estimation method to estimate $\underset{\sim}{\mu}$ and Σ.
Determine the first revised estimates.